



UMS
UNIVERSITI MALAYSIA SABAH

INDUSTRIAL TRAINING HANDBOOK

**FACULTY OF ENGINEERING
UNIVERSITI MALAYSIA SABAH**

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SECTION 1 INTRODUCTION

1.1 Introduction

Industrial Training is a compulsory 5-credit hour course for students who have completed their 6 semesters of study in the Faculty of Engineering (FKJ). Students are required to undergo industrial training for a minimum period of 10 weeks at the relevant industry or government body. Upon completion, student will be graded with **Pass with Distinction (Lulus dengan Cemerlang)**, **Pass (Lulus)** or **Fail (Gagal)** to indicate the level of achievement in industrial training. However, the grade will not contribute to the Cumulative Grade Point Average (CGPA).

1.2 Course Objectives

The objectives of the industrial training are:

- i. To provide opportunities for students to apply their knowledge in a real world environment
- ii. To expose students to an industrial working environment

1.3 Course Outcomes

By the end of the course, the student will demonstrate:

- i. Initiative and responsibility in carrying out assigned tasks
- ii. An ability to write technical documents / reports related to the work completed
- iii. An ability to creatively utilize technical resources both from prior relevant course work as well from sources students must seek out on their own
- iv. An ability to articulate their critical thinking

SECTION 2 MANAGEMENT OF INDUSTRIAL TRAINING

2.1 Status and Credit Hour

This course carries 5-credit hour and compulsory for students who have completed their 6 semesters of study in the Faculty of Engineering (FKJ). The course assessment is as follows:

Marks	Assessment
0 - 49	Fail
50 - 89	Pass
90 - 100	Pass with Distinction

2.2 Duration

- The minimum duration of the Industrial Training is 10 weeks.
- In case the company requested for longer duration, the approval of extension request will depend on the decision of Industrial Training Committee.

2.3 Industrial Training Procedure

The industrial training procedures are divided into three stages; before the industrial training, during the industrial training and after the industrial training. These procedures are elaborated in details in the next section.

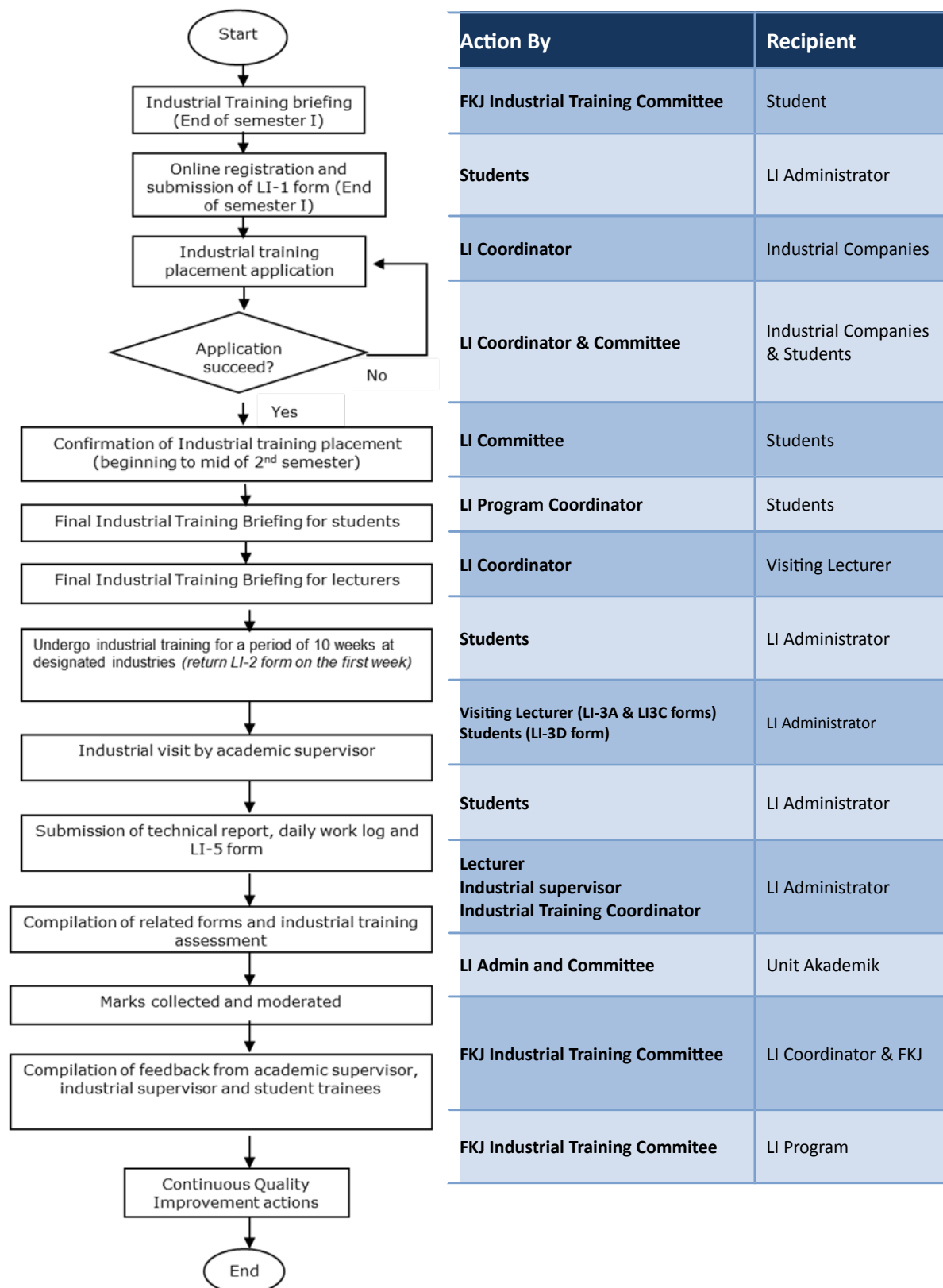


Figure 1 Flowchart for Industrial Training procedure.

2.3.1 Before the Industrial Training

- i. Student must complete the Student Registration Form (LI - 1) online.
- ii. Student holds responsibility to get placement at relevant industry or government body. The FKJ Industrial Training Committee will monitor all applications for placement.
- iii. Once confirmed, student is not allowed to change placement.
- iv. Student must attend the Industrial Training briefings conducted by the Industrial Training Committee.

2.3.2 During the Industrial Training

- i. Student and Industrial Supervisor must complete the Industrial Training Commencement Form (LI – 2 can be found in this Industrial Training Handbook 2017) when reporting for duty. The student and industrial supervisor must discuss and produce an Industrial Training Plan for the period of training. The completed form and Industrial Training Plan must be submitted by the student to the Industrial Training Administration within 7 days from the date of industrial training commencement.
- ii. Student must prepare an A4 size hardcover log book to record daily activities. Please refer to Appendix B for further details. The Industrial Supervisor must certify this log book daily or weekly.
- iii. Industrial Training Visit:
 - The Industrial Training Committee will arrange the date in which the Visiting Lecturer will conduct a visit to the organization.
 - The student must prepare and present the log book, training schedule and Forms LI-3A and LI-3C to the Visiting Lecturer during the visit.
 - If the visit is not possible, a telephone interview or video conferencing must be arranged between the lecturer, student and industrial supervisor.
- iv. At week 8th of training, student holds the responsibility to give LI-3B and LI-4B forms to be completed by the Industrial Supervisor.

2.3.3 After the Industrial Training

- i. Student must submit the Industrial Training Technical Report, log book, sealed envelopes (LI-3B and LI-4B), LI-4A and LI-5 to FKJ Industrial Training Administration on the first Monday of the following semester.
- ii. If the documents are incomplete, marks will be deducted from the final assessment.

SECTION 3 ROLES AND RESPONSIBILITY

3.1 Industrial Supervisor

An Industrial Supervisor is an officer in the organization where the student is attached and person who will be supervising the work of the student during the attachment. The Industrial Supervisor is preferably an engineer or equivalent. The proposed duties include;

- i. Supervising student's activities.
- ii. Discussing with the Industrial Training Coordinator on matters relevant to the training, projects and student's work performance.
- iii. Evaluate and assess student's performance during the training period, and to submit the LI-3B and LI-4B form at the end of the industrial training.

3.2 Visiting Lecturer

The Visiting Lecturer appointed by the FKJ to visit or to contact the student during the industrial training period. The appointment is based on the expertise and the experience of the lecturer related to the industry. The visit may not be carried out for very established companies i.e. Intel, JKR, Telekom etc. or companies that have shown to provide proper training scheme to students from previous visit(s). The duties of the Visiting Lecturer include:

- i. To attend briefing given by the industrial Training coordinator regarding the visit to the industries.
- ii. To visit the student to study the suitability of the working environment.
- iii. To discuss with the Industrial Supervisor on the student performance and any other matters related to Industrial Training.
- iv. To meet with higher management to establish industrial linkages.
- v. To fill in the LI-3A and LI-3C forms during the visit.
- vi. To submit all the forms to Industrial Training Administration within 15 days upon returning to UMS.
- vii. To evaluate the student's performance based on the Technical report and Daily Work Log Book.

3.3 Student

Student is expected to perform the duties that include:

- i. Student must confirm which organization to report to before starting their Industrial Training.
- ii. Student should report for duty to the Industrial Supervisor assigned by the organization and agreed by the Faculty.
- iii. Student must understand and adhere to the rules and regulations of the organization they are placed in.
- iv. Student must always wear appropriate attire during the industrial training according to the dress code of the organization.
- v. Student must know how to manage their time efficiently and follow the working hours set by the organization.
- vi. Student must put on their best behavior in line with the university's guidelines at all times while undergoing the Industrial Training.

- vii. Student must submit medical certificate (MC) to the Industrial Supervisor if not able to come to work.
- viii. Student must uphold the UMS image.

SECTION 4 EVALUATION SYSTEM

The student will be assessed according to the following criteria:

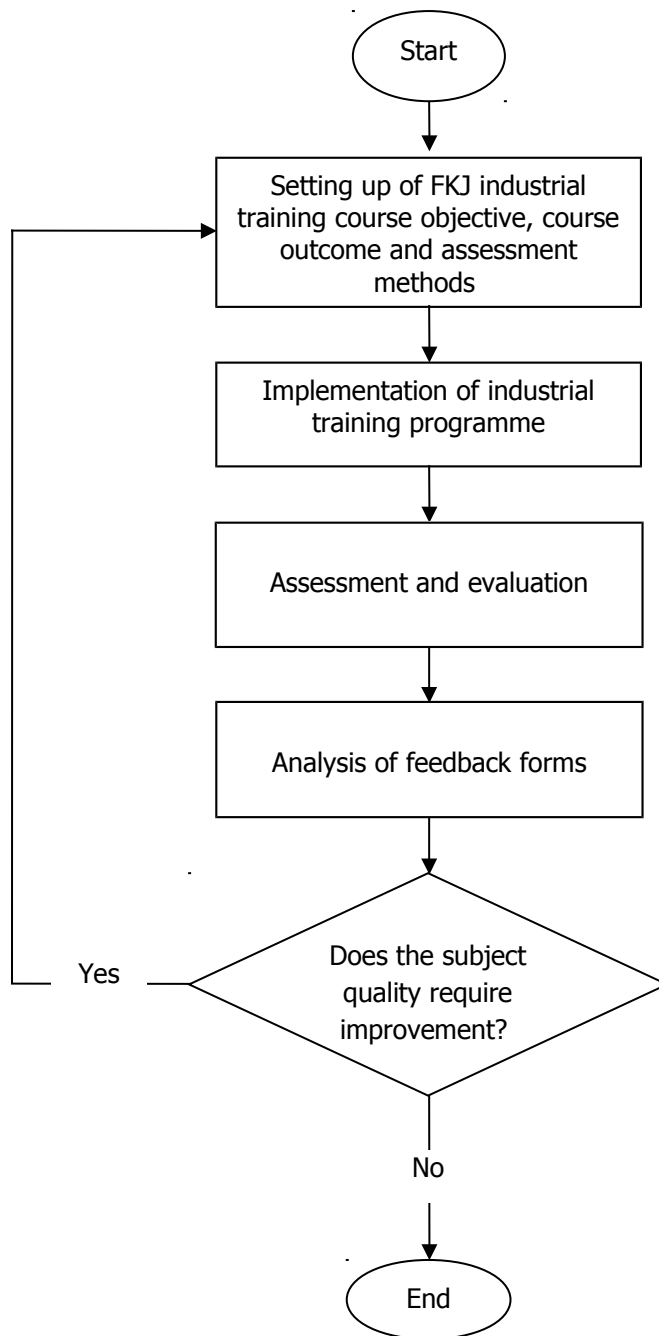
i. Academic Supervisor (50%)

ITEM	DISTRIBUTION %	MARKS CALCULATION
Log Book (Format & Content) i. Daily tasks, ii. New knowledge, iii. Problem-solving (Define, Identify, solve, verify) iv. Trouble-shooting (If applicable)	20%	MARKS/5* 20
Technical Report Understanding: • Formulation/Justification/Methodology	10%	MARKS/ 5* 10
Technical Report Contribution : • Analysis/Results and Discussions/Conclusions	20%	MARKS/ 5* 20

ii. Industrial Supervisor (50%)

ITEM	DISTRIBUTION %	MARKS CALCULATION
Application of knowledge	20%	MARKS/5 * 20
Communication skills	10%	MARKS/5 * 10
Responsibility and integrity	10%	MARKS/5 * 10
Initiative and willingness to learn	10%	MARKS/5 * 10

SECTION 5 CONTINUOUS QUALITY IMPROVEMENT (CQI)



SECTION 6 FREQUENTLY ASKED QUESTION(S) (FAQs)

1. What steps should student take to find companies?

- Firstly students need to fill and submit the LI-1 form and 3 company addresses to the Industrial Training Administrator.
- Secondly the industrial training committee will apply for industrial training places for the students.
- Thirdly the company will reply to the Faculty.
- Finally the industrial training committee will inform the student.
- Refer to flowchart for the flow of industrial training procedure.

2. Will industrial training students receive allowance?

- Usually the students do not receive industrial training allowance. However some companies choose give allowance to the students.

3. Are industrial training students insured? If yes, who is the point of contact?

- Yes. The point of contact is the Penolong Pendaftar HEP. Please call +6 088-320000 ext.: 3117 and ask to be connected to the penolong pendaftar kanan HEP.

4. Can industrial training students apply for a short leave/long leave?

- Yes, with replacement.

5. Can industrial training students take leave from industrial training to attend the PALAPES / SUKSIS program?

- Yes. But the student needs to apply leave from the company and inform the Faculty with the attached letter from the PALAPES / SUKSIS program. The students are advised to replace the two weeks spent on the PALAPES / SUKSIS program. The two weeks can be replaced by working extra hours for the rest of the 8 weeks of their industrial training period. The students are not allowed replace the two weeks with semester time.

6. What is the format of the technical report?

- The format for the technical report can be found in the Industrial Training Handbook 2017 (Refer to Appendix A)

7. When is the industrial training report submission due?

- The technical report should be submitted on the first Monday of the following semester. Failing to submit on time may result in failure of industrial training program.

8. What should we do during our industrial training?

- You should discuss with your industrial supervisor regarding your industrial training activities and schedule. You have to adhere with the company's rules and regulations, and be safe at all times. Remember you represent the Universiti Malaysia Sabah with your best behavior.

SECTION 7 APPENDICES

Appendix A – Industrial Training Technical Report Format Guideline

FACULTY OF ENGINEERING

Elements that should be present:

Executive Summary

Title

Acknowledgement

Table of Contents

Chapter 1 : Introduction (Minimum 4 pages)

- Brief Background of the Organization
 - Organizational History, Structure and Organization Chart, Product/Service, Type of Business, and Human Resources
- Workflow
 - Activities or workflow within the department/unit/section where the student is undergoing Industrial Training
- Objectives of the Student Activities
 - Job scope and objectives of the Industrial Training in that organization in the specified area

Chapter 2 : Job training, experience and accomplishments (Minimum 2 pages)

- Detailed work experience during Industrial Training in essay form.

Chapter 3 : Project Activities (Minimum 20 pages)

- Details of the (minimum 1 activity, more is accepted) activities/projects undertaken by the student during the Industrial Training according to the followings headings:
 - Location
 - Procedure/Methods accompanied by:
 - For Engineers – relevant diagrams, drawings and/or hand sketches containing sufficient details to enable draughtsman to work them up into drawings without further guidance.
 - For Information Technologists - relevant software development, network management and troubleshooting, system analysis and design etc.
 - Result/Achievement
 - Conclusion

Chapter 4 : Critical Analysis (Minimum 2 pages)

- Weakness/strength relating to the job/training/suitability/problem (if any)

Chapter 5 : Suggestion/Resolution & Conclusion (Minimum 2 pages)

- How to solve the problem, any comments about the training and the organization and conclusion

References/Bibliography

Appendix

Attachment

Instructions to the authors:

1. Use Times New Roman font size 11 and 1.5 spacing.
2. Page numbers should be positioned at the top on the right hand-side of the page.
3. Use A4 size paper.
4. Number of pages should be between 30-50 including figures and photos.
5. All tables and figures must be labeled.

Appendix B - Instructions on Writing Industrial Training Log Book

A log book is one of the written reports required for Industrial Training assessment. By nature, it should be HANDWRITTEN in a LOG BOOK. The log book should comprise the following items:

1. Student and Industrial Training Information (Typed on A4 paper and pasted onto the first page of the ledger book)
2. Industrial training plan - Outline of training and proposals (Handwritten)
3. Daily record (Handwritten)
 - a. Date
 - b. time
 - c. Activities:
 - i. Topic / Activity / Issue / Problem
 - ii. Objective / Goal / Purpose
 - iii. Finding / Solution / Conclusion
 - iv. Others
 - d. Verification from Person in charge

Log Book: Industrial Training Plan

Students are required to discuss with industrial supervisor on training schemes during the training period. It may be in a form of training schedule on different departments, skills or mini projects. The training schedule can also be illustrated in the form of Gantt chart if relevant. Students are obligated to comply with the industrial supervisor's instructions.

Log Book: Daily Report

Comprise of detail activities on daily basis. Students need to state the date and time of the activities and tasks that are relevant to industrial training only as detailed above.

Sample of student and industry information **On page 1 of log book**

Student Information

Full Name : _____

Matrix No : _____ IC No. : _____

Program : _____

Contact No. : _____ E-Mail : _____

Permanent Address : _____

Next of kin (relationship): _____ Phone / HP : _____

Current Address: _____

In case of emergency, please contact : _____

Company Information

Company : _____
Supervisor : _____
Post : _____ Phone/ HP : _____
Company Address : _____

Period of Industrial Training : _____ until _____

Faculty Contact Detail

Faculty Level Chief Industrial

Training Coordinator : _____
Contact Number : _____

Program Level Industrial

Training Coordinator : _____
Program : _____
Contact Number : _____

Example of Industrial Training Plan
On page 2 of ledger book

Industrial Training Plan

Period	Tasks / Activities
04 – 10 July 2016	Department: Human Resource Activities: 1. Familiarize with the new working environment. 2. Meeting with industrial supervisor. 3. Briefing about company rules and regulations. 4. Awareness on safety.
13 – 20 July 2016	Department: Maintenance Activities:

Example of Daily Report
On page 3 of ledger book
Date:

Topic/Activity/Issue/Problem:

- 1. Boiler leakage (technical)*
- 2. Attending monthly meeting (management)*

**To be elaborated in details.*

Objective/Goal/Purpose:

- 1. To identify the leakage problems and rectify the problems. (technical)*
Include sketch, diagram, etc., if applicable.
- 2. Issue raised up in meeting (management)*

**To be elaborated in details.*

Finding/Solution/Conclusion:

- 1. Finding: Low pressure in piping system, Solution: New pipe is required.*
Include sketch, diagram, etc., if applicable.
- 2. The decisions during meeting (management)*

**To be elaborated in details.*

Others:

Verified by:

(Signature and name)

Appendix C - Forms

FORMS NO.	ITEMS
LI – 1	Student Registration Form

LI – 2	Industrial Training commencement Form
LI – 3A	Industrial Training Visit Form (FKJ Lecturer Report)
LI – 3B	Industrial Training visit Form (Industrial Supervisor Report)
LI – 3C	Industrial Linkage & Collaboration Questionnaire (Industry Feedback)
LI – 3D	Industrial Training Visit Form (Student Feedback)
LI – 3E	FKJ, UMS Alumni's Employer Survey
LI – 4A	Industrial Training Evaluation Form (Academic Supervisor Report)
LI – 4B	Industrial Training Evaluation Form (Industrial Supervisor Report)
LI – 5	Receipt of Industrial Training Report
LI - 6	Leave application form